

Every sufficiently large primorial multiple is an equal fair-die size

Research proof; independent outline reviews by two collaborators

30 September 2026

Fix $n \geq 1$ and put $P_n = \prod_{p \leq n} p$, the empty product being one. A positive word on n letters is fair with equal count M if each letter occurs M times and every ordering of every subset S occurs as a subsequence exactly $M^{|S|}/|S|!$ times. The condition on all n -letter orders implies these subset conditions by marginalization, so this is precisely the usual notion of equal-sized fair dice.

Theorem 1. *For each n there is an integer N_n such that equal-sized n -fair dice with M faces per die exist for every $M \geq N_n$ divisible by P_n . Conversely, every feasible equal count is divisible by P_n . More specifically, there is a feasible L for which $L+P_n$ is also feasible. Thus, writing $a = L/P_n$, every $P_n k$ with $k \geq a(a-1)$ is feasible.*

The prefix-completion step is a direct refinement of the manuscript's already stated symmetrization lemma (credited there to Purcell). The new consequence here comes from combining this completion property with positive right fractions and the signed-integrality characterization.

The proof is constructive, but the present construction gives no useful bound of the form $N_n = \exp(O(n))$. It therefore settles eventual arithmetic feasibility, not the manuscript's shortest-word question.

The reduced signature algebra

Let $\mathcal{A}_{\mathbb{Q}}$ have basis the empty word and all words of distinct letters $X_1, \dots, X_n$. Products concatenate words with disjoint letter sets and are zero otherwise. Its positive-degree ideal is nilpotent. The signature of a positive word $i_1 \cdots i_s$ is

$$\text{sig}(i_1 \cdots i_s) = \prod_{j=1}^s (1 + X_{i_j}).$$

Its coefficients count subsequences with distinct letters. Set $S = X_1 + \cdots + X_n$ and $T_M = \exp(MS)$. Equal-count fairness is exactly $\text{sig}(w) = T_M$. In particular, $T_M T_N = T_{M+N}$. Let $\mathcal{S}$ be the monoid of positive signatures, including the empty word, and let G be the group it generates. The inverse of $1 + X_i$ is $1 - X_i$. Every element of G has a finite logarithm which is a Lie polynomial in the X_i , by the finite BCH formula.

Lemma 2 (Vanishing invariant Lie polynomials). *If a homogeneous Lie polynomial L of degree $k > 1$ in $\mathcal{A}_{\mathbb{Q}}$ is invariant under all permutations of the n letters, then $L = 0$.*

Proof. Every degree- k associative word with distinct letters is in one orbit under the permutations. Thus L must have the same coefficient b on every such word. Map $\mathcal{A}_{\mathbb{Q}}$ to the commutative algebra $\mathbb{Q}[x_1, \dots, x_n]/(x_1^2, \dots, x_n^2)$. Every Lie polynomial of degree greater than one maps to zero. But the coefficient of each squarefree degree- k monomial in the image of L is $k!b$. Consequently $b = 0$. $\square$

Lemma 3 (Universal positive prefix completion). *Every positive word u is a literal prefix of an equal-count fair word. If no letter occurs more than B times in u , a completion with count $B(n!)^{n-1}$ per letter exists. For the empty word one may choose any positive B .*

Proof. Append letters until every count is B , obtaining a word w with $\log \text{sig}(w) = BS +$ terms of degree at least two. Suppose inductively that a positive word w has

$$\log \text{sig}(w) = CS + L_k + \text{terms of degree greater than } k, \quad k \geq 2,$$

where all degrees $2, \dots, k-1$ vanish. Concatenate the $n!$ relabelings $\sigma(w)$, with the identity permutation first. In the BCH logarithm of this product, the degree-one term is $n!CS$ and the degree- k term is $\sum_{\sigma \in \mathfrak{S}_n} \sigma(L_k)$. Indeed, the degree-one terms are all proportional to S and commute, while any bracket involving a degree- k term has degree at least $k+1$. The degree- k sum vanishes by Lemma 2. The new word therefore has no defect through degree k , and retains w as a literal prefix. Apply this operation for $k = 2, \dots, n$. There are no degrees above n in the algebra, so the resulting signature is $T_{B(n!)^{n-1}}$. $\square$

Lemma 4 (Clearing a signed signature by a fair word). *For every $g \in G$ there is a positive integer L with $T_L \in \mathcal{S}$ and $gT_L \in \mathcal{S}$.*

Proof. First, any $u, v \in \mathcal{S}$ have a common right multiple in $\mathcal{S}$. By Lemma 3, write $uu' = T_A$ and $vv' = T_B$, with positive A, B and $u', v' \in \mathcal{S}$. Then

$$u(u'T_B) = v(v'T_A) = T_{A+B}.$$

It follows directly that every element of G is a right fraction ab^{-1} , where $a, b \in \mathcal{S}$. For completeness, right fractions contain all generators and inverses. They are closed under products: if $bx = cy$ is a common right multiple, then

$$(ab^{-1})(cd^{-1}) = ax(dy)^{-1}.$$

Now complete b to a fair signature, writing $bb' = T_L$ with $b' \in \mathcal{S}$ and positive L . Then $gT_L = ab' \in \mathcal{S}$. $\square$

The arithmetic input and conclusion

We use the following signed-signature theorem proved separately in `size_bounds/signed_fair_signatures.tex`:

$$\exp\left(\sum_i c_i X_i\right) \in G \iff |I|! \mid \prod_{i \in I} c_i \quad (\emptyset \neq I \subseteq [n]), \quad c_i \in \mathbb{Z}.$$

For clarity, its sufficiency is elementary: correct homogeneous errors successively in the integral multilinear Lie basis $[X_{i_1}, [\dots, [X_{i_{k-1}}, X_a]] \dots]$, whose coefficients are read from the associative words ending in X_a . Nested group commutators realize $1 + B$ for these basis brackets exactly, because products repeating a letter vanish. This writes every integral fair signature as a signed word.

For equal $c_i = M$, the divisibility conditions hold precisely when $P_n \mid M$. Necessity follows by taking a subset of size p for every prime $p \leq n$. Conversely, if $p \mid M$, then $v_p(k!) \leq k \leq kv_p(M)$ for every $k \leq n$; primes exceeding n do not divide $k!$. In particular $T_{P_n} \in G$.

Apply Lemma 4 to $g = T_{P_n}$. Both T_L and T_{L+P_n} are positive fair signatures. Necessarily $P_n \mid L$. Let $a = L/P_n$. Concatenation realizes all nonnegative integer combinations of the consecutive integers a and $a + 1$, multiplied by P_n . Every integer $k \geq a(a - 1)$ is such a combination: writing $k = qa + r$ with $0 \leq r < a$, one has $q \geq a - 1 \geq r$, and hence $k = (q - r)a + r(a + 1)$. This proves Theorem 1.

What the proof does and does not use. No general nilpotent-semigroup saturation theorem is needed. The special ingredient is the vanishing of

permutation-invariant multilinear Lie errors, which gives literal positive prefix completion. The noncentrality of the fair ray causes no problem after establishing the common-right-multiple property; one never commutes a fair block past an arbitrary signed word.

Independent checks. Two collaborators independently checked the proof outline, including the invariant-Lie argument and the order of multiplication in the right-fraction identity. The companion exact script verifies the completion construction for 20 inputs with $1 \leq n \leq 4$, using actual positive words, testing the literal prefix and every subset-order count. All checks passed.